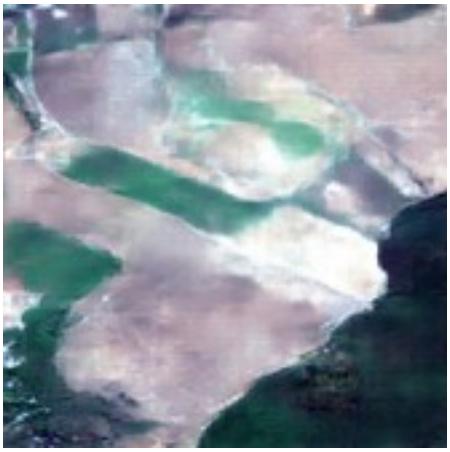
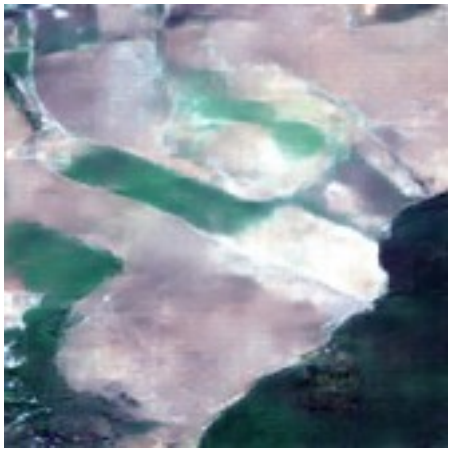
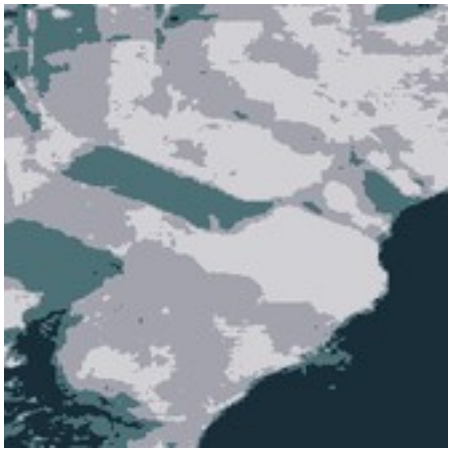
SAR Image Analysis Report



Original SAR (not found; preview shown)



Colorized



K-Means Segmented

Metadata:

- Original File: Not available
- Colorized File: colorized_1764152957.png
- Processed On: 2025-11-26 15:59:32 IST
- SAR Classifier Confidence: 0.000
- Colorization Model: Pix2Pix GAN
- KMeans Clusters: 4
- Colorized Resolution: 256 x 256 px

Cluster Analysis:

Cluster 0: Likely Water (Blue Tone) | Mean RGB: (214, 214, 214) | Area: 25.31%

Cluster 1: Likely Water (Blue Tone) | Mean RGB: (80, 115, 121) | Area: 15.75%

Cluster 2: Likely Water (Blue Tone) | Mean RGB: (169, 167, 178) | Area: 40.03%

Cluster 3: Mixed / Other Surface | Mean RGB: (25, 46, 57) | Area: 18.91%

Land Cover Summary:

- Likely Water (Blue Tone) → 25.31%
- Likely Water (Blue Tone) → 15.75%
- Likely Water (Blue Tone) → 40.03%
- Mixed / Other Surface → 18.91%

Note: Colorized image is GAN-generated and for visualization only.

Analysis Details & Recommendations

Interpretation Notes:

The area is dominated by likely water (blue tone) (81.09% combined).

Secondary presence: mixed / other surface (18.91%).

K-Means groups pixels by color similarity — labels are heuristic.

Consider adding multispectral or time-series data for improved analysis.

Recommendations:

The analysis shows dominant land cover is water (35.15%). Significant water presence suggests focusing on hydrological monitoring and flood risk assessment for the area. Secondary clusters show distributed variations; using time-series imagery would help confirm seasonal changes. Recommended actions: collect ground-truth samples where possible, prioritize monitoring based on dominant class, and re-run analysis after incorporating higher-resolution data or additional bands.

Confidence Indicators:

SAR Detection Confidence: 0.000
Dominant Class Percentage: 40.03%
Classification Reliability: High

Limitations:

- KMeans is unsupervised and may not match exact land-cover classes.
- GAN colorization is synthetic and does not represent true spectral values.
- Classification depends on image quality and may contain uncertainty.
- Single-image analysis; use multi-temporal data for trend-based conclusions.

Final Summary:

The analysis indicates predominant vegetation with mixed surfaces.

No major water bodies or dense urban zones detected in this image.

Recommend periodic monitoring and ground-truth sampling to improve confidence.